Week 2 | Solve With Instructors

Q1) Given the kernel matrix

$$\mathbf{K} = \begin{bmatrix} 1 & 0.8 & 0.5 \\ 0.8 & 1 & 0.6 \\ 0.5 & 0.6 & 1 \end{bmatrix}$$

Find the centered kernel matrix.

Hint: For any matrix $\mathbf{X}_{d \times n}$, the one way to get the mean vector is:

$$\frac{1}{n}\mathbf{X}\mathbf{1}$$

Here $\mathbf{1}$ is vector of order n having all entries as 1.

First try to find how to center $\phi(\mathbf{X})$ using above hint and apply that idea to $\mathbf{K} = \phi(\mathbf{X})^T \phi(\mathbf{X})$.

Solution:

From $\frac{1}{n}\mathbf{X1}$, we get the mean vector but to center the dataset we have to subtract this mean vector from each data point. This operation can be done like this:

$$\mathbf{X}_{\textbf{centered}} = \mathbf{X} - \mathbf{X1}_{n\times n}$$

Note here $\mathbf{1}_{n\times n}$ is $\frac{1}{n}\begin{bmatrix} & \vdots & \\ \cdots & 1 & \cdots \\ & \vdots & \end{bmatrix}$

The operation $\mathbf{X1}_{n\times n}$ will gives us a matrix of mean vector that is each column of this matrix represents the mean of $\mathbf{X}$ data matrix.

Kernel matrix is represented as

$$\mathbf{K} = \phi(\mathbf{X})^T \phi(\mathbf{X})$$

The centered matrix of $\phi(\mathbf{X})$ is $\phi(\mathbf{X}) - \phi(\mathbf{X})\mathbf{1}_{n\times n}$. With this let's compute centered kernel matrix form

$$\begin{aligned}
\mathbf{K}_c &= \phi(\mathbf{X})_c^T \phi(\mathbf{X})_c \\
&= [\phi(\mathbf{X}) - \phi(\mathbf{X})\mathbf{1}_{n\times n}]^T [\phi(\mathbf{X}) - \phi(\mathbf{X})\mathbf{1}_{n\times n}] \\
&= \underbrace{\phi(\mathbf{X})^T \phi(\mathbf{X})} \; - \; \underbrace{\phi(\mathbf{X})^{\mathbf{T}} \phi(\mathbf{X})}\mathbf{1}_{n\times n} \; - \; \mathbf{1}_{n\times n}\underbrace{\phi(\mathbf{X})^T \phi(\mathbf{X})} \; + \; \mathbf{1}_{\mathbf{n}\times\mathbf{n}}\underbrace{\boldsymbol{\phi}(\mathbf{X})^{\mathbf{T}} \boldsymbol{\phi}(\mathbf{X})}\mathbf{1}_{n\times n} \\
&= \mathbf{K} - \mathbf{K}\mathbf{1}_{n\times n} - \mathbf{1}_{n\times n}\mathbf{K} + \mathbf{1}_{n\times n}\mathbf{K}\mathbf{1}_{n\times n}
\end{aligned}$$

Given Kernel matrix is

$$\mathbf{K} = \begin{bmatrix} 1 & 0.8 & 0.5 \\ 0.8 & 1 & 0.6 \\ 0.5 & 0.6 & 1 \end{bmatrix}$$

Using the result that we got before

$$\mathbf{K} - \mathbf{K}\mathbf{1}_{n\times n} - \mathbf{1}_{n\times n}\mathbf{K} + \mathbf{1}_{n\times n}\mathbf{K}\mathbf{1}_{n\times n}$$

$$= \begin{bmatrix} 1 & 0.8 & 0.5 \\ 0.8 & 1 & 0.6 \\ 0.5 & 0.6 & 1 \end{bmatrix} - \begin{bmatrix} 0.76 & 0.76 & 0.76 \\ 0.8 & 0.8 & 0.8 \\ 0.7 & 0.7 & 0.7 \end{bmatrix} - \begin{bmatrix} 0.76 & 0.8 & 0.7 \\ 0.76 & 0.8 & 0.7 \\ 0.76 & 0.8 & 0.7 \end{bmatrix} + \begin{bmatrix} 0.75 & 0.75 & 0.75 \\ 0.75 & 0.75 & 0.75 \\ 0.75 & 0.75 & 0.75 \end{bmatrix}$$

$$= \begin{bmatrix} 0.22 & -0.01 & -0.21 \\ -0.01 & 0.16 & -0.14 \\ -0.21 & -0.14 & 0.36 \end{bmatrix}$$

Q2) Consider the following kernel

$$k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$$
$$k(\mathbf{x}, \mathbf{y}) = \left(\mathbf{x}^T \mathbf{y}\right)^2$$

where $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$. Which of the following transformation ϕ may corresponds to k?

- ○ $\phi[x_1, x_2]^T = [x_1, x_1 x_2, x_2]^T$
- ○ $\phi[x_1, x_2]^T = \left[x_1^2, x_1 + x_2, x_2^2\right]^T$
- ○ $\phi[x_1, x_2]^T = \left[x_1, \sqrt{2} x_1^2 x_2^2, x_2\right]^T$
- ○ $\phi[x_1, x_2]^T = \left[x_1^2, \sqrt{2} x_1 x_2, x_2^2\right]^T$

Solution:

On expanding this $\left(\mathbf{x}^T\mathbf{y}\right)^2$

$$\begin{aligned} k(\mathbf{x},\mathbf{y}) &= \left(\mathbf{x}^T\mathbf{y}\right)^2 \\ &= (x_1y_1 + x_2y_2)^2 \\ &= x_1^2y_1^2 + 2x_1y_1x_2y_2 + x_2^2y_2^2 \end{aligned}$$

It can be written as

$$\begin{bmatrix} x_1^2 & \sqrt{2}x_1x_2 & x_2^2 \end{bmatrix} \begin{bmatrix} y_1^2 \\ \sqrt{2}y_1y_2 \\ y_2^2 \end{bmatrix}$$

Hence, the transformation ϕ is

$$\phi[x_1, x_2]^T = \left[x_1^2, \sqrt{2}x_1x_2, x_2^2\right]^T$$

Q3) A function k is defined as

$$k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$$
$$k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1\mathbf{x}_2 + \mathbf{x}_1^2\mathbf{x}_2^2 + \mathbf{x}_1^3\mathbf{x}_2 + 1$$

Is k a valid kernel?

◯ Yes
◯ No

Solution:

From Mercer's theorem, we know that a valid kernel is symmetric. Let's check for the given function

$$k(\mathbf{x}_1, \mathbf{x}_2) = \mathbf{x}_1\mathbf{x}_2 + \mathbf{x}_1^2\mathbf{x}_2^2 + \mathbf{x}_1^3\mathbf{x}_2 + 1 \tag{1}$$

$$k(\mathbf{x}_2, \mathbf{x}_1) = \mathbf{x}_2\mathbf{x}_1 + \mathbf{x}_2^2\mathbf{x}_1^2 + \mathbf{x}_2^3\mathbf{x}_1 + 1 \tag{2}$$

We can see that the above two equations are not same. Hence, the given function is not a valid kernel function.

Q4) Suppose $k_1(\mathbf{x}, \mathbf{y}) = \exp\left(-\|\mathbf{x} - \mathbf{y}\|^2\right)$ is a Gaussian (RBF) kernel and $k_2(\mathbf{x}, \mathbf{y}) = \left(\mathbf{x}^T\mathbf{y} + 1\right)^3$ is a polynomial kernel of degree 3. If a new kernel function k is defined as :

$$k(\mathbf{x}, \mathbf{y}) = k_1(\mathbf{x}, \mathbf{y}) + 3k_2(\mathbf{x}, \mathbf{y})$$

then find the kernel matrix $\mathbf{K}$ for the following dataset:

$$\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \end{bmatrix}\right\}$$

Solution:

The new kernel is defined as

$$k(\mathbf{x}, \mathbf{y}) = k_1(\mathbf{x}, \mathbf{y}) + 3k_2(\mathbf{x}, \mathbf{y})$$

Let we name data points as

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \ \mathbf{x}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \ \mathbf{x}_3 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \ \mathbf{x}_4 = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

Let's calculate value for each $k(x_i, x_j)$ from the above definition. Here we will only show calculation for two data points.

$$\begin{aligned} k(\mathbf{x}_1, \mathbf{x}_1) &= k_1(\mathbf{x}_1, \mathbf{x}_1) + 3k_2(\mathbf{x}_1, \mathbf{x}_1) \\ &= \exp\left(-\|\mathbf{x}_1 - \mathbf{x}_1\|^2\right) + 3\left(\mathbf{x}_1^T \mathbf{x}_1 + 1\right)^3 \\ &= \exp(0) + 3(5+1)^3 \\ &= 1 + 3 \times 216 \\ &= 649 \end{aligned}$$

$$\begin{aligned} k(\mathbf{x}_2, \mathbf{x}_4) &= k_1(\mathbf{x}_2, \mathbf{x}_4) + 3k_2(\mathbf{x}_2, \mathbf{x}_4) \\ &= \exp\left(-\|\mathbf{x}_2 - \mathbf{x}_4\|^2\right) + 3\left(\mathbf{x}_2^T \mathbf{x}_4 + 1\right)^3 \\ &= \exp\left(-(4)^2\right) + 3(-3+1)^3 \\ &= \exp(-16) + 3(-8) \\ &\approx -24 \end{aligned}$$

$$k(\mathbf{x}_1, \mathbf{x}_1) = 649 \qquad k(\mathbf{x}_1, \mathbf{x}_2) = 192.01 \qquad k(\mathbf{x}_1, \mathbf{x}_3) = -24 \qquad k(\mathbf{x}_1, \mathbf{x}_4) = -192$$

$$k(\mathbf{x}_2, \mathbf{x}_1) = 192.01 \qquad k(\mathbf{x}_2, \mathbf{x}_2) = 649 \qquad k(\mathbf{x}_2, \mathbf{x}_3) = -192 \qquad k(\mathbf{x}_2, \mathbf{x}_4) = -24$$

$$k(\mathbf{x}_3, \mathbf{x}_1) = -24 \qquad k(\mathbf{x}_3, \mathbf{x}_2) = -192 \qquad k(\mathbf{x}_3, \mathbf{x}_3) = 649 \qquad k(\mathbf{x}_3, \mathbf{x}_4) = 192.01$$

$$k(\mathbf{x}_4, \mathbf{x}_1) = -192 \qquad k(\mathbf{x}_4, \mathbf{x}_2) = -24 \qquad k(\mathbf{x}_4, \mathbf{x}_3) = 192.01 \qquad k(\mathbf{x}_4, \mathbf{x}_4) = 649$$

The $\mathbf{K}$ matrix would be

$$\begin{bmatrix} 649 & 192.01 & -24 & -192 \\ 192.01 & 649 & -192 & -24 \\ -24 & -192 & 649 & 192.01 \\ -192 & -24 & 192.01 & 649 \end{bmatrix}$$

$Q5)$ There are n data points (mean of the data points is 0). P is a 3×3 matrix. $P_{ij} = x_i^T x_j$.

1. How many data points are there?

2. If the eigenvalues of P are l_1, l_2, l_3 ($l_1 \geqslant l_2 \geqslant l_3 > 0$), the trace of the covariance matrix is given by :

   a. $l_1 + l_2 + l_3$

   b. $\frac{1}{n}(l_1 + l_2 + l_3)$

   c. $\frac{1}{n^2}(l_1 + l_2 + l_3)$

   d. None of these

3. If the eigenvalues of P are $15, 30$ and 24, then find the variance along the second principal component.

Hint $b)$: Trace of a square matrix is given by the sum of its diagonal elements. The trace of a square matrix is also given by the sum of all the eigenvalues.

ANSWERS

1. If $P_{ij} = x_i^T x_j$, then $P = X^T X$, where every column of x is a data point. $X^T X$ is an $n \times n$ matrix. Therefore $n = 3$.

$$X = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix}$$

2. The dataset is mean-centered.

   The non-zero eigenvalues of XX^T and $X^T X$ are the same. Here, we can see that $X^T X$ has three non-zero eigenvalues. Therefore, XX^T will also have three non-zero eigenvalues. This would mean that the covariance matrix will also have exactly three non-zero eigenvalues.

   Let the non-zero eigenvalues of the covariance matrix be $\lambda_1, \lambda_2, \lambda_3$, where $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3 > 0$.

   The non-zero eigenvalues of XX^T are l_1, l_2, l_3.

   Covariance matrix for a mean-centered dataset is $\frac{1}{n} XX^T$. If the non-zero eigenvalues of XX^T is l_1, l_2 and l_3, then, the non-zero eigenvalues of the covariance matrix are $\frac{l_1}{n}, \frac{l_2}{n}, \frac{l_3}{n}$. Therefore, $\lambda_1 = \frac{l_1}{n}, \lambda_2 = \frac{l_2}{n}, \lambda_3 = \frac{l_3}{n}$.

   The trace of the covariance matrix is $\lambda_1 + \lambda_2 + \lambda_3 = \frac{1}{n}(l_1 + l_2 + l_3)$.

3. The eigenvalues of $X^T X$ are $15, 30, 24$. The non-zero eigenvalues of XX^T are $15, 30, 24$. The non-zero eigenvalues of the covariance matrix are $5, 10$ and 8. (refer to the answer of question (b).

   The second largest eigenvalue of the covariance matrix is the variance along the second principal component. Therefore, the variance along the second principal component is 8.

$Q6)$ Every element in a dataset D is a vector of $\mathbb{R}^3$.
$k : D \times D \rightarrow \mathbb{R}$ is defined as follows :

$$k(x, x') = \left(x^T x' + 1\right)^2$$

Every data point in a dataset is an element of $\mathbb{R}^2$. Every column vector is a data point. Using $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$, we map every data point to $\mathbb{R}^6$.

The following observation is made :

$$\phi(x)^T \phi(y) = k(x, y)$$

where $k(x, y) = \left(x^T y + 1\right)^2$ for every x and y in the data set

Every data point in the mapped space is in the nullspace of vector u, where

$$u = \left(\frac{1}{a^2}, \frac{-\sqrt{2}h}{a^2}, \frac{1}{b^2}, \frac{-\sqrt{2}k}{b^2}, 0, \frac{h^2}{a^2} + \frac{y^2}{k^2} - r^2\right)^{\mathbf{T}}, \ a, b, k, h \in \mathbb{R}$$

a. What could possibly be $\phi(x)$, if $x = (x_1, x_2)$
b. What could possibly be the relationship between the features of every data point in $\mathbb{R}^2$

ANSWER

The given kernel is as follows

$$k(x,y) = \left(x^T y + 1\right)^2$$

Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$

$$k(x,y) = (x_1y_1 + x_2y_2 + 1)^2 = x_1^2y_1^2 + 2x_1x_2y_1y_2 + 2x_1y_1 + x_2^2y_2^2 + 2x_2y_2 + 1$$

We can write this as :

$$\left(x_1^2, \sqrt{2}x_1, x_2^2, \sqrt{2}x_2, \sqrt{2}x_1x_2, 1\right)\begin{pmatrix} y_1^2 \\ \sqrt{2}y_1 \\ y_2^2 \\ \sqrt{2}y_1 \\ \sqrt{2}y_1y_2 \\ 1 \end{pmatrix}$$

Therefore, $\phi(x) = \left(x_1^2, \sqrt{2}x_1, x_2^2, \sqrt{2}x_2, \sqrt{2}x_1x_2, 1\right)^{\mathbf{T}}$

It is given that $\phi(x)^T u = 0$, where $u = \left(\frac{1}{a^2}, \frac{-\sqrt{2}h}{a^2}, \frac{1}{b^2}, \frac{-\sqrt{2}k}{b^2}, 0, \frac{h^2}{a^2} + \frac{y^2}{k^2} - r^2\right)^{\mathbf{T}}$.

$$\left(x_1^2, \sqrt{2}x_1, x_2^2, \sqrt{2}x_2, \sqrt{2}x_1x_2, 1\right) \times \left(\frac{1}{a^2}, \frac{-\sqrt{2}h}{a^2}, \frac{1}{b^2}, \frac{-\sqrt{2}k}{b^2}, 0, \frac{h^2}{a^2} + \frac{y^2}{k^2} - r^2\right)^{\mathbf{T}} = 0$$

This turns out to be :

$$\frac{x_1^2}{a^2} + \frac{h^2}{a^2} - \frac{2x_1 h}{a^2} + \frac{x_2^2}{b^2} + \frac{k^2}{b^2} - \frac{2x_2 k}{b^2} - 1 = 0$$

This can be written as

$$\frac{(x_1 - h)^2}{a^2} + \frac{(y_1 - k)^2}{b^2} = 1$$

This is the equation of an ellipse.

$Q7)$ We have n data points in $\mathbb{R}^d$. It is observed that $d \gg n$. We use the following algorithm to find the principal components. There might be some errors below. Find if any.

1. Put every data point in the columns of a matrix and call it X
2. Find $K = XX^T$
3. Find the non-zero eigenvalues of K (let there be k of them) : $l_1 \geqslant l_2 \geqslant ... \geqslant l_k > 0$. Their corresponding eigenvectors are $e_1, e_2,..., e_k$
4. The ith principal component is given by $\dfrac{Xe_i}{l_i}$.

- In the second step, we have to mean-center the dataset
- $K = X^T X$ and not XX^T
- The ith principal component is given by $\dfrac{Xe_i}{\sqrt{l_i}}$.

$Q8)$ Every data point of a dataset is a vector in $\mathbb{R}^2$. Every (x_1, x_2) in the dataset satisfies the following :

$$\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} = 1$$

Give a mapping $\phi : \mathbb{R}^2 \to \mathbb{R}^6$ defined as

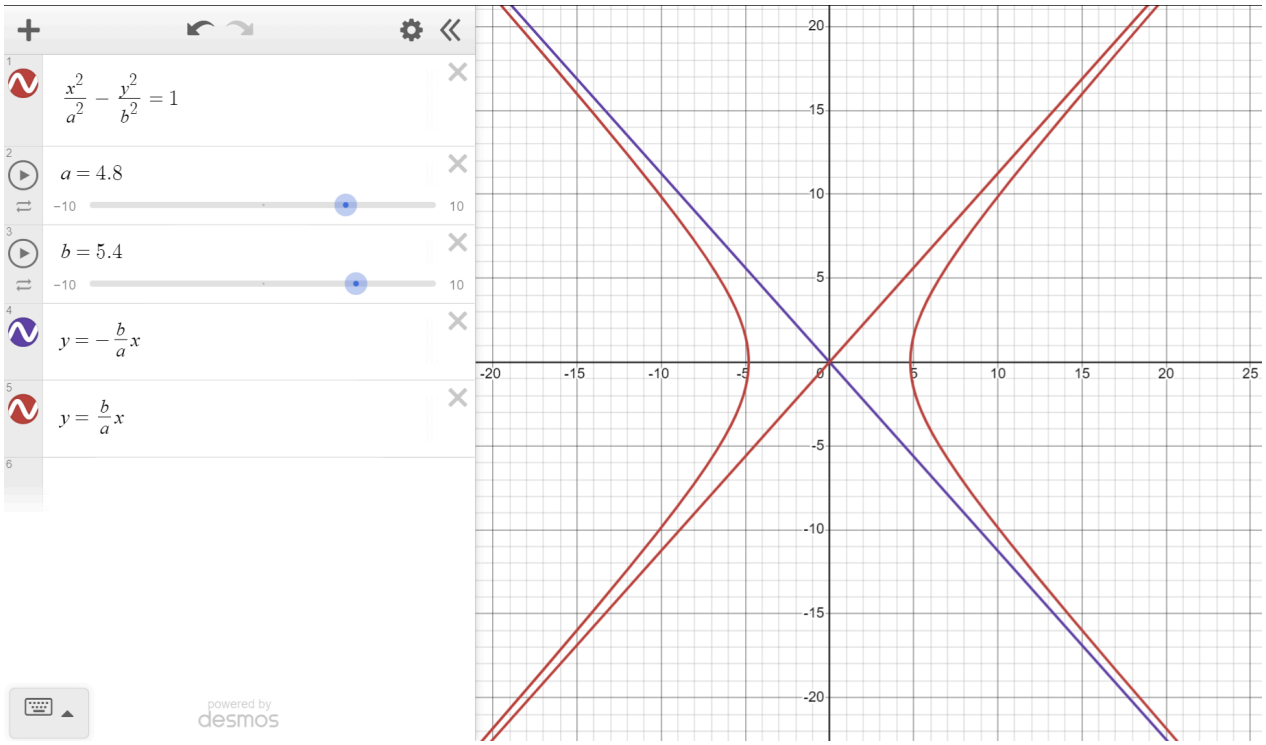
$$\phi(x) = \begin{pmatrix} x_1^2 & \sqrt{2}x_1x_2 & \sqrt{2}x_1 & x_2^2 & \sqrt{2}x_2 & 1 \end{pmatrix}^T$$

such that $\phi(x)^T\phi(y)$ is given by the polynomial kernel of degree 2

.

Find a vector which is orthogonal to every data point in the mapped space.

Additional information. The given equation represents a hyperbola with vertices at $(a, 0)$ and $(-a, 0)$ and with $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$ being oblique asymptotes.



ANSWER

Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$. Polynomial kernel of degree 2 is given by

$$k(x, y) = \left(x^T y + 1\right)^2$$

This can be written as follows

$$\begin{pmatrix} x_1^2 & \sqrt{2}x_1x_2 & \sqrt{2}x_1 & x_2^2 & \sqrt{2}x_2 & 1 \end{pmatrix} \begin{pmatrix} y_1^2 \\ \sqrt{2}y_1y_2 \\ \sqrt{2}y_1 \\ y_2^2 \\ \sqrt{2}y_2 \\ 1 \end{pmatrix}$$

$\phi(x) = \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2}x_1 \\ x_2^2 \\ \sqrt{2}x_2 \\ 1 \end{pmatrix}$ is the mapping.

Every data point satisfies

$$\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} = 1$$

$$\frac{x_1^2}{a^2} - \frac{x_2^2}{b^2} - 1 = 0$$

This can be written as

$$\left(x_1^2, \quad \sqrt{2}x_1x_2, \quad \sqrt{2}x_1, \quad x_2^2, \quad \sqrt{2}x_2, \quad 1 \right) \cdot \left(\frac{1}{a^2}, \quad 0, \quad 0, \quad \frac{-1}{b^2}, \quad 0, -1 \right) = 0$$

Every data point in the mapped space is orthogonal to

$$\left(\frac{1}{a^2}, \quad 0, \quad 0, \quad \frac{-1}{b^2}, \quad 0, -1 \right)$$

Question 9

Consider the dataset as shown in the figure. We want to project the dataset into another feature space so that it lives in a linear subspace of a higher-dimensional space. We apply the kernel PCA with a polynomial kernel of the appropriate degree to achieve the same. What will be the dimension of the transformed feature space?

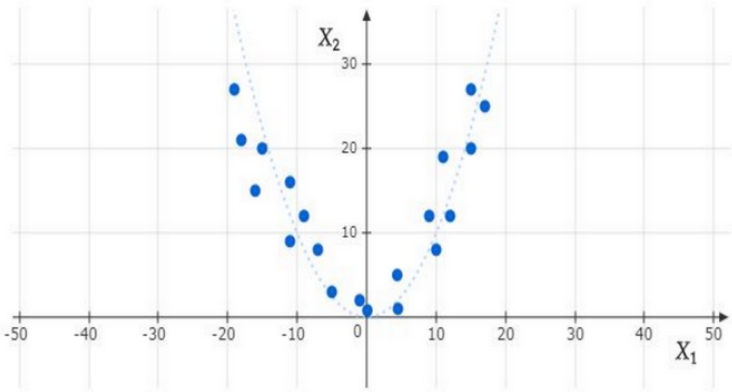


Figure 1: Dataset figure

Question 10

Consider the following kernel function:

$$k : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$$

$$k\left((x_1, x_2)^T, (y_1, y_2)^T\right) = 1 + x_1y_1 + x_2y_2 + x_1^2y_1^2 + x_2^2y_2^2 + x_1y_1x_2y_2$$

Find the appropriate transformation mapping ϕ for the given kernel

Question 11

A dataset containing 200 examples in three-dimensional space has been transformed into a higher-dimensional space using a polynomial kernel of degree two. What will be the dimension of the transformed feature space?

Question 12

Consider that 1000 data points belonging to d-dimensional space have a non-linear relationship. We apply kernel PCA to reduce the dimension of the data points and take the first k principal components. Can the value of k be larger than d?

Solution 9

We are given a dataset that appears to follow a quadratic relationship in the (X_1, X_2) plane. The goal is to project this dataset into a higher-dimensional feature space using Kernel PCA with a polynomial kernel of appropriate degree, such that the data becomes linearly separable in the new space. 1. The dataset follows a quadratic relationship, likely of the form:

$$X_2 = X_1^2$$

2. The appropriate kernel to capture this quadratic pattern is the polynomial kernel of degree $p = 2$.

Polynomial Kernel

The polynomial kernel of degree p is given by:

$$K(X_1, X_2) = (X_1 \cdot X_2 + 1)^p$$

For this problem, since the relationship is quadratic, we set $d = 2$.

Transformed Feature Space Dimension

Using a polynomial kernel of degree 2, we map the data into a new feature space where each feature is a combination of the original features up to degree 2. These combinations are:

$$\{1, X_1, X_2, X_1^2, X_2^2, X_1X_2\}$$

Thus, the total number of features in the transformed space is 6.

Solution 10

The given kernel function is:

$$k\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = 1 + x_1y_1 + x_2y_2 + x_1^2y_1^2 + x_2^2y_2^2 + x_1x_2y_1y_2$$

We aim to find the transformation mapping ϕ such that:

$$k\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = \langle \phi(x_1, x_2), \phi(y_1, y_2) \rangle$$

From the kernel function, we can identify that the corresponding mapping ϕ should include all combinations of the features x_1 and x_2 and their squares, as well as the constant term 1.

The transformation ϕ is:

$$\phi(x_1, x_2) = \big(1, x_1, x_2, x_1^2, x_2^2, x_1x_2\big)$$

and similarly:

$$\phi(y_1, y_2) = \big(1, y_1, y_2, y_1^2, y_2^2, y_1y_2\big)$$

Now, the dot product $\langle \phi(x_1, x_2), \phi(y_1, y_2) \rangle$ in the transformed feature space is:

$$\langle \phi(x_1, x_2), \phi(y_1, y_2) \rangle = 1 \cdot 1 + x_1 \cdot y_1 + x_2 \cdot y_2 + x_1^2 \cdot y_1^2 + x_2^2 \cdot y_2^2 + (x_1x_2) \cdot (y_1y_2)$$

This results in:

$$\langle \phi(x_1, x_2), \phi(y_1, y_2) \rangle = 1 + x_1y_1 + x_2y_2 + x_1^2y_1^2 + x_2^2y_2^2 + x_1x_2y_1y_2$$

Solution 11

We are given a dataset with 200 examples in 3-dimensional space, and we apply a polynomial kernel of degree 2.

The dimension of the transformed feature space for a polynomial kernel of degree d is given by the binomial coefficient:

$$\text{Dimension} = \binom{n+d}{d}$$

where n is the number of original features (in this case, $n = 3$) and d is the degree of the polynomial kernel (here, $d = 2$).

Substitute $n = 3$ and $d = 2$ into the formula:

$$\text{Dimension} = \binom{3+2}{2} = \frac{(3+2)!}{3!2!}$$

Simplifying the factorials:

$$\text{Dimension} = \frac{5!}{3!2!} = \frac{120}{6 \times 2} = 10$$

The dimension of the transformed feature space is: 10

Solution 12

No, the value of k cannot be larger than d when considering standard PCA in the original linear space. In linear PCA, the maximum number of principal components is bounded by the smaller value between the number of data points n and the dimensionality of the data d. In other words:

$$k \leq \min(n, d)$$

However, Kernel PCA is different from linear PCA because it maps the original data into a higher-dimensional feature space using a kernel function. This allows Kernel PCA to capture non-linear relationships between the data points. In the feature space, the effective dimensionality could be much higher than d.

As a result, in Kernel PCA, the number of principal components k can exceed d, as the feature space often has a much higher dimensionality than the original space. The number of principal components is still bounded by the number of data points n:

$$k \leq n$$

Therefore, in Kernel PCA, k can indeed be larger than d, but it will always be less than or equal to n, the number of data points.